

Two-Stage Hard-Constrained WMU Placement for Fault Detection and Zone Localization under IBR-Induced Oscillatory Backgrounds

Hjin

Department of Electrical and Electronic Engineering
Hanyang University

Email: 271123352+hjiny@users.noreply.github.com

Abstract—The increasing penetration of inverter-based resources (IBRs) introduces oscillatory background conditions that can complicate waveform-based fault monitoring. At the same time, non-fault switching events such as load switching should not trigger protection-oriented fault decisions. This paper proposes a two-stage waveform measurement unit (WMU) placement framework that separates fault detection from fault-zone localization. Stage 1 uses a hard-constrained detection anchor WMU and requires zero normal false positives, zero LoadSwitch false positives, zero single-line-to-ground fault false negatives, and zero three-phase fault false negatives. Stage 2 is activated only after a fault decision and localizes the fault to a predefined topology zone using a physically interpretable current-disturbance argmax rule. A modified IEEE 30-bus deterministic waveform dataset with IBR-like 25 Hz subsynchronous oscillation background, normal cases, 5%, 15%, and 30% LoadSwitch cases, and SLG/three-phase fault cases is used for evaluation. The detection stage selects Bus 27 as a one-WMU anchor with zero evaluated detection errors and a fault-score margin of 0.511616. Exhaustive subset search for localization shows that no subset with $k \leq 7$ WMUs achieves zero zone error, while five zero-error subsets exist at $k = 8$. The selected final localization placement is {2, 11, 13, 16, 20, 23, 24, 27}, which includes the Bus 27 detection anchor and achieves zero zone-localization error over the 60 evaluated fault cases.

Index Terms—Waveform measurement unit, WMU placement, inverter-based resources, subsynchronous oscillation, load switching, fault detection, zone localization, hard constraint, IEEE 30-bus system.

I. Introduction

The transition from synchronous-generator-dominated systems to inverter-based-resource (IBR)-rich systems has introduced new monitoring and protection challenges. Unlike conventional rotating machines, IBRs interact with the grid through converter controls and synchronization loops. This interaction can produce oscillatory behavior in weak grids, especially when converter control dynamics, network impedance, and operating conditions align unfavorably. Prior studies on weak-grid IBRs and wind-energy converters have reported subsynchronous oscillation (SSO), subsynchronous control interaction, and impedance-driven stability phenomena [1]–[10]. These studies imply that future fault monitoring should not assume a clean pre-event waveform.

Waveform measurement units (WMUs) provide synchronized high-resolution voltage and current waveforms. Compared with conventional phasor measurement units (PMUs), WMUs can preserve transient shapes, time-domain changes, spectral components, and phase-dependent waveform signatures. Recent synchro-waveform studies have used waveform data to locate transient events, detect incipient faults, classify power-quality events, and identify fault locations or types [11]–[13]. However, if installing WMUs at every bus is not economical, the placement problem becomes central.

Classical PMU placement studies commonly formulate the problem as the minimization of sensor count subject to complete or partial network observability [14]–[18]. Fault-aware PMU placement extends this idea toward fault observability and fault location [19], [20]. These formulations provide an important measurement-placement foundation, but most conventional observability formulations do not directly enforce waveform-based discrimination between faults and non-fault switching events under IBR-like oscillatory backgrounds.

This distinction is important because load switching, capacitor switching, transformer energization, and other switching events are frequently treated as non-fault disturbances in protection and transient-classification literature [21]–[30]. A placement that maximizes average classification accuracy may still be unsuitable if it misclassifies a non-fault switching event as a fault. Accordingly, this paper treats LoadSwitch cases as non-fault events and imposes an explicit zero-LoadSwitch-false-positive constraint.

The research objective is to determine a minimum WMU placement architecture that (i) discriminates fault events from normal and LoadSwitch events under IBR-like SSO background conditions and (ii) localizes detected faults to predefined topology zones. The proposed result is deliberately framed as a deterministic in-dataset case-study result under a specified feature set, topology-zone definition, and decision rule. It is not a claim of universal deployment-ready optimality.

The main contributions are as follows.

- 1) A two-stage WMU placement framework is formulated

to separate fault detection from zone localization.

- 2) The detection stage imposes hard protection-oriented constraints: NormalFP=0, LoadSwitchFP=0, SLGFN=0, and ThreePhaseFN=0.
- 3) The localization stage uses a physically interpretable $dI_energy_{3ph,max}$ argmax rule over an installed WMU set.
- 4) Exhaustive subset search proves that the first zero-zone-error localization solutions occur at $k = 8$ under the specified formulation.
- 5) A final installed set $\{2, 11, 13, 16, 20, 23, 24, 27\}$ is selected because it satisfies the localization constraint while retaining Bus 27 as the detection anchor.

II. Proposed Two-Stage Framework

Fig. 1 summarizes the proposed two-stage framework. The first stage decides whether an event is a fault or non-fault event. The second stage is activated only for detected faults and estimates the topology zone of the fault.

Let $\mathcal{B} = \{1, 2, \dots, 30\}$ denote the IEEE 30-bus candidate bus set. Each event case i is assigned a binary label

$$y_i \in \{0, 1\}, \quad (1)$$

where $y_i = 0$ denotes a non-fault case, including normal operation and LoadSwitch disturbances, and $y_i = 1$ denotes a fault case, including SLG and three-phase faults. For fault cases, the true faulted bus is denoted by b_i , and the corresponding topology zone is

$$z_i = Z(b_i), \quad (2)$$

where $Z(\cdot)$ maps each bus to its predefined topology zone.

The design motivation is that the WMU subset that is sufficient for fault/non-fault detection is not necessarily sufficient for location identification. The final system therefore uses

$$\mathcal{S}_{det} = \{27\} \quad (3)$$

for detection and

$$\mathcal{S}_{loc} = \{2, 11, 13, 16, 20, 23, 24, 27\} \quad (4)$$

for localization.

III. Stage 1: Hard-Constrained Fault Detection

A. Event Label Definition

Let e_i denote the event type of case i . The non-fault and fault event sets are defined as

$$\mathcal{C}_0 = \{\text{SSO_Normal}, \text{SSO_LoadSwitch}\}, \quad (5)$$

$$\mathcal{C}_1 = \{\text{SSO_SLG_Fault}, \text{SSO_ThreePhase_Fault}\}. \quad (6)$$

The binary fault label is

$$y_i = \begin{cases} 0, & e_i \in \mathcal{C}_0, \\ 1, & e_i \in \mathcal{C}_1. \end{cases} \quad (7)$$

This binary mapping follows protection-oriented transient studies in which load switching is treated as a normal or non-fault disturbance rather than as a fault.

B. Bus-Wise Normalization and Evidence Features

Let $x_{i,b,f}$ be the raw value of feature f for case i at observed bus b . For each bus and feature, the 5th and 95th percentiles are computed as

$$q_{b,f}^{0.05} = Q_{0.05}(\{x_{i,b,f} : i \in \mathcal{I}\}), \quad (8)$$

$$q_{b,f}^{0.95} = Q_{0.95}(\{x_{i,b,f} : i \in \mathcal{I}\}). \quad (9)$$

The normalized feature is

$$\tilde{x}_{i,b,f} = \text{clip}_{[0,1]} \left(\frac{x_{i,b,f} - q_{b,f}^{0.05}}{\max(q_{b,f}^{0.95} - q_{b,f}^{0.05}, \epsilon)} \right), \quad \epsilon = 10^{-12}. \quad (10)$$

This normalization is performed per observed bus, not globally, to prevent buses with naturally larger absolute waveform magnitudes from dominating the placement score.

The fault evidence feature set includes voltage/current disturbance energy, voltage sag, RMS-current jump, sequence/unbalance ratios, and apparent-impedance change features. In compact form,

$$\mathcal{F}_{fault} = \{f_1^{fault}, f_2^{fault}, \dots, f_{13}^{fault}\}. \quad (11)$$

These features reflect voltage sag, current disturbance, sequence imbalance, and apparent-impedance change.

LoadSwitch evidence is based on the 67–77 Hz voltage cycle-difference ratio in three phases:

$$\mathcal{F}_{LS} = \{\text{dV_E72_ratio_A}, \text{dV_E72_ratio_B}, \text{dV_E72_ratio_C}\}. \quad (12)$$

C. Detection Score and Hard Constraints

For each case and bus, fault evidence is defined as

$$F_{i,b} = \max_{f \in \mathcal{F}_{fault}} \tilde{x}_{i,b,f}, \quad (13)$$

and LoadSwitch evidence is defined as

$$L_{i,b} = \min_{\phi \in \{A,B,C\}} \tilde{x}_{i,b,\text{dV_E72_ratio}_\phi}. \quad (14)$$

For the detection anchor b_d , the score is

$$s_{i,b_d} = F_{i,b_d} - \lambda L_{i,b_d}, \quad \lambda = 0.5. \quad (15)$$

The maximum non-fault score and minimum fault score are

$$M_0(b_d) = \max_{i:y_i=0} s_{i,b_d}, \quad m_1(b_d) = \min_{i:y_i=1} s_{i,b_d}. \quad (16)$$

The separation margin is

$$\gamma_{det}(b_d) = m_1(b_d) - M_0(b_d). \quad (17)$$

If $\gamma_{det}(b_d) > 0$, the midpoint threshold is selected:

$$\tau(b_d) = \frac{M_0(b_d) + m_1(b_d)}{2}. \quad (18)$$

The predicted label is

$$\hat{y}_i = \mathbb{I}(s_{i,b_d} > \tau). \quad (19)$$

Fig. 1. Proposed two-stage framework

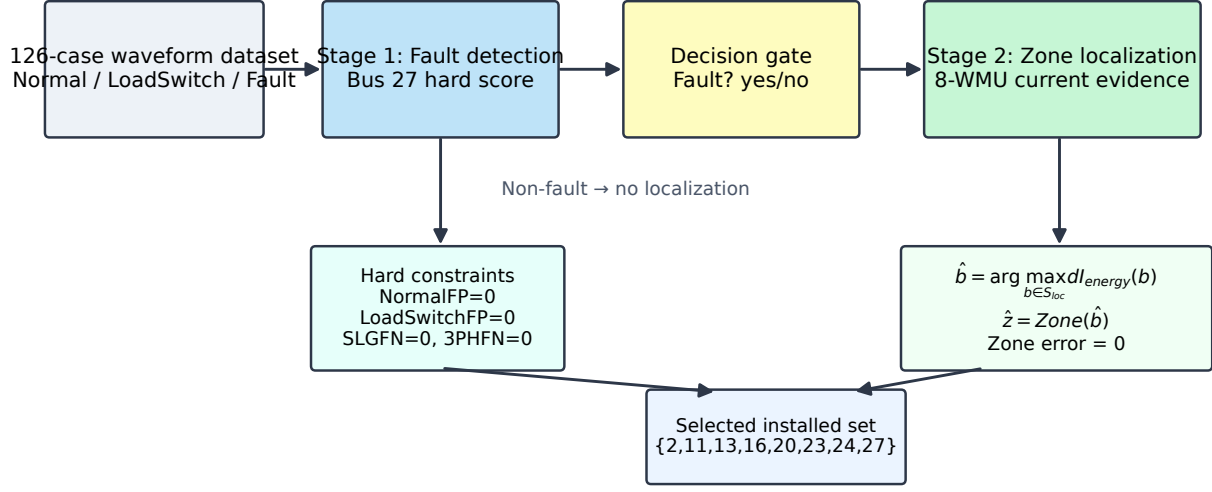


Fig. 1. Proposed two-stage framework. Stage 1 performs hard-constrained fault detection using Bus 27 as the detection anchor. Stage 2 performs current-disturbance-based zone localization using the selected eight-WMU localization set.

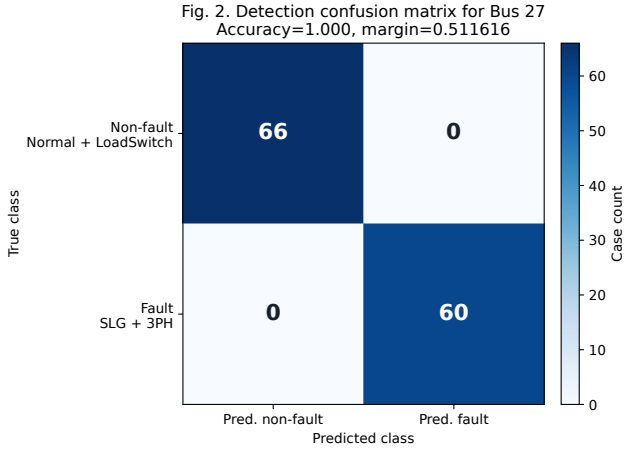


Fig. 2. Detection confusion matrix for Bus 27. The evaluated dataset contains 66 non-fault cases and 60 fault cases, and Bus 27 produces zero evaluated detection errors.

A detection anchor is feasible only if

$$\begin{aligned} \text{NormalFP} &= 0, & \text{LoadSwitchFP} &= 0, \\ \text{SLGFN} &= 0, & \text{ThreePhaseFN} &= 0. \end{aligned} \quad (20)$$

Bus 27 satisfies these hard constraints and provides the detection confusion matrix in Fig. 2.

IV. Stage 2: Current-Disturbance-Based Zone Localization

A. Argmax Localization Rule

After a case is classified as a fault, Stage 2 estimates the faulted topology zone. For each fault case i and installed

Fig. 6. Current disturbance argmax explanation

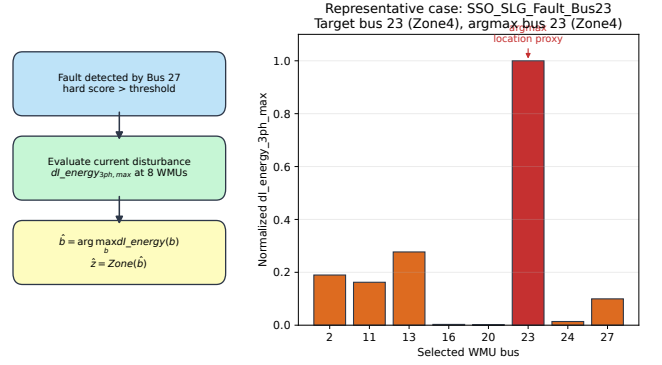


Fig. 3. Current disturbance argmax explanation. After a fault is detected, the localization stage compares $dI_{\text{energy}}_{3ph,max}$ across the selected WMU buses, chooses the maximum as the location proxy, and maps the proxy bus to a topology zone.

localization WMU $b \in S_{loc}$, the localization stage uses

$$D_{i,b} = dI_{\text{energy}}_{3ph,max}(i, b). \quad (21)$$

The predicted location proxy bus is selected by

$$\hat{b}_i(S_{loc}) = \arg \max_{b \in S_{loc}} D_{i,b}. \quad (22)$$

The predicted fault zone is then

$$\hat{z}_i(S_{loc}) = Z(\hat{b}_i(S_{loc})). \quad (23)$$

This rule is physically interpretable: the installed WMU with the largest current-disturbance energy is treated as a proxy for the faulted region. Fig. 3 illustrates this decision rule.

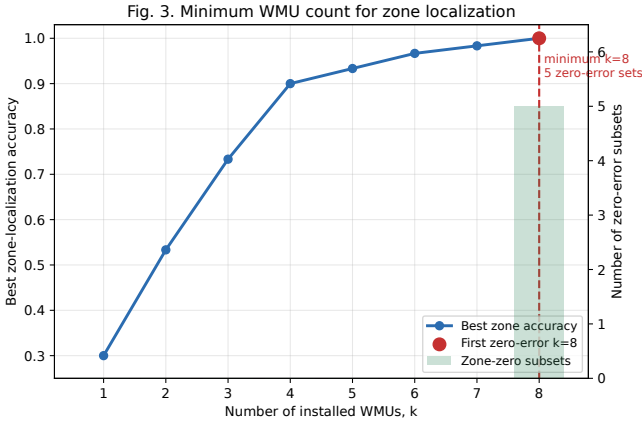


Fig. 4. Minimum WMU count for zone localization. Exhaustive subset search shows that the first zero-zone-error solutions occur at $k = 8$.

B. Zone-Localization Constraint

The zone-localization error for a localization set S_{loc} is defined over the evaluated fault-case set $\mathcal{F}$:

$$\text{ZoneErr}(S_{loc}) = \sum_{i \in \mathcal{F}} \mathbb{I}(\hat{z}_i(S_{loc}) \neq z_i). \quad (24)$$

A localization set is feasible if

$$\text{ZoneErr}(S_{loc}) = 0. \quad (25)$$

The minimum zone-localization placement problem is

$$S_{loc}^* = \arg \min_{S \subseteq \mathcal{B}} |S| \quad (26)$$

subject to

$$\text{ZoneErr}(S) = 0. \quad (27)$$

The term minimum is therefore defined with respect to the specified dataset, zone definition, current-disturbance feature, and argmax rule.

C. Exhaustive Search

The exhaustive search evaluates all candidate WMU subsets in increasing order of cardinality. For each subset size k , all combinations $S \subseteq \mathcal{B}$ satisfying $|S| = k$ are tested. The search terminates at the first k for which at least one subset satisfies zero zone error. The search result is shown in Fig. 4. No zero-error subset exists for $k = 1, 2, \dots, 7$. At $k = 8$, five zero-error subsets are found among 5,852,925 evaluated combinations. Thus,

$$k_{zone}^* = 8. \quad (28)$$

Among the five feasible eight-WMU subsets, this paper selects

$$S_{loc}^* = \{2, 11, 13, 16, 20, 23, 24, 27\}. \quad (29)$$

This set is selected because it satisfies the zero zone-localization constraint and includes Bus 27, the detection anchor identified in Stage 1. Fig. 5 shows the selected placement on the IEEE 30-bus topology, and Fig. 6 shows the zone-localization confusion matrix.

Fig. 4. Selected 8-WMU placement on IEEE 30-bus topology

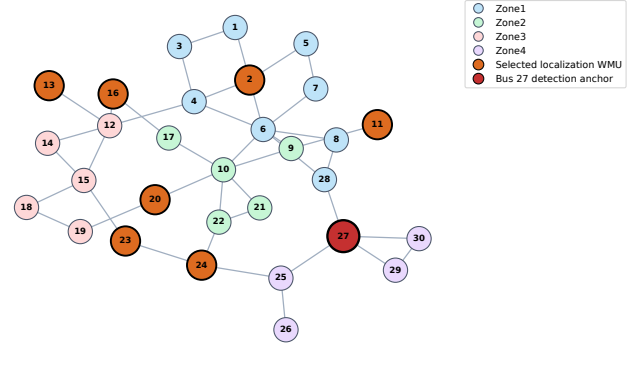


Fig. 5. Selected eight-WMU placement on the IEEE 30-bus topology. The final localization set is $\{2, 11, 13, 16, 20, 23, 24, 27\}$, and Bus 27 is also the detection anchor.

Fig. 5. Zone localization confusion matrix

Selected set $\{2, 11, 13, 16, 20, 23, 24, 27\}$; accuracy=1.000

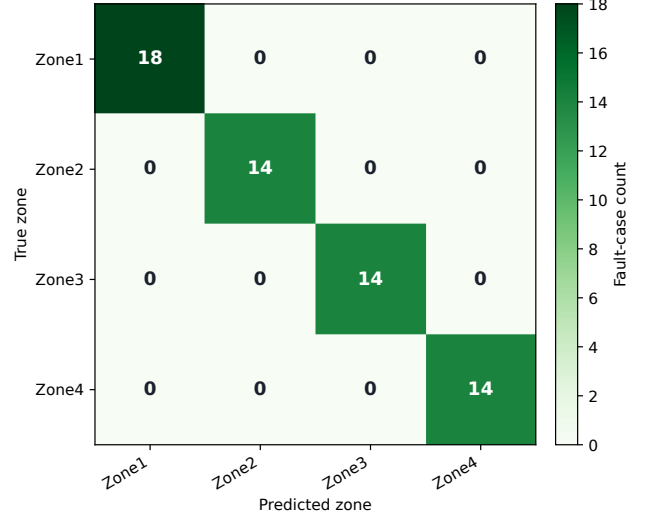


Fig. 6. Zone localization confusion matrix for the selected eight-WMU set. The selected placement achieves zero zone-localization error over the 60 evaluated fault cases.

V. Case Study

A. Test System and Dataset

The case study uses a modified IEEE 30-bus system. An IBR-like 25 Hz SSO background is included in all event classes to reflect oscillatory monitoring conditions associated with weak-grid IBR interactions. The dataset contains 126 deterministic simulation cases, as summarized in Table I.

The dataset includes 66 non-fault cases and 60 fault cases. The LoadSwitch cases are divided into 5%, 15%, and 30% magnitudes to test whether a selected placement remains feasible as non-fault switching severity changes.

B. Detection Result

The Bus 27 detection anchor yields zero evaluated false positives and zero evaluated false negatives. Table II

TABLE I
Combined Dataset Composition

Event type	Magnitude	Cases	Binary class
SSO_Normal	–	3	Non-fault
SSO_LoadSwitch	5%	21	Non-fault
SSO_LoadSwitch	15%	21	Non-fault
SSO_LoadSwitch	30%	21	Non-fault
SSO_SLG_Fault	–	30	Fault
SSO_ThreePhase_Fault	–	30	Fault
Total	–	126	–

TABLE II
Selected Bus 27 Detection Performance

Metric	Value
Selected detection WMU	Bus 27
NormalFP	0
LoadSwitchFP	0
SLGFN	0
ThreePhaseFN	0
TotalFP / TotalFN	0 / 0
Binary accuracy	1.000
Non-fault max score	-0.038596
Fault min score	0.473020
Threshold	0.217212
Fault-score margin	0.511616

summarizes the result. The score separation satisfies

$$-0.038596 < 0.217212 < 0.473020, \quad (30)$$

where -0.038596 is the largest non-fault score, 0.473020 is the smallest fault score, and 0.217212 is the threshold. The resulting margin is 0.511616 .

C. Localization Result

The localization exhaustive search result is summarized in Table III. The best zone accuracy increases with k , but a zero-zone-error subset does not exist until $k = 8$.

The selected localization set $\{2, 11, 13, 16, 20, 23, 24, 27\}$ achieves zero zone-localization error over all evaluated fault cases. Because the set includes the Bus 27 detection anchor, the final two-stage installation does not require a separate detection-only sensor outside the localization placement.

VI. Discussion

The proposed result highlights an important distinction between fault detection and fault localization. Under the evaluated hard constraints, a single WMU at Bus 27 is sufficient for binary fault detection. However, localizing the detected fault to the correct topology zone requires a larger sensor set. The exhaustive search shows that no subset with $k \leq 7$ can satisfy zero zone error under the specified feature, zone definition, and argmax rule. The first feasible localization solutions occur at $k = 8$.

This distinction prevents an overstatement of the detection-only result. A single WMU should not be claimed to solve all monitoring objectives merely because

TABLE III
Exhaustive Zone-Localization Search Summary

k	Total subsets	Zero-error subsets	Best accuracy
1	30	0	0.5000
2	435	0	0.7167
3	4,060	0	0.8333
4	27,405	0	0.9000
5	142,506	0	0.9500
6	593,775	0	0.9667
7	2,035,800	0	0.9833
8	5,852,925	5	1.0000

it can separate fault and non-fault cases in the evaluated dataset. Instead, the proposed two-stage architecture explicitly quantifies the additional WMU requirement needed to support fault-zone identification.

The result should also be interpreted with caution. The dataset is deterministic and limited in operating-condition diversity. Normal cases are limited, stochastic load variation is not included, and the SSO background is fixed. Measurement noise, missing channels, communication delay, synchronization error, fault resistance variation, inception-angle variation, and other uncertainties should be included in future validation.

VII. Conclusion

This paper proposed a two-stage hard-constrained WMU placement framework for fault detection and zone localization under IBR-like oscillatory background conditions. Stage 1 uses Bus 27 as a detection anchor and achieves zero evaluated NormalFP, LoadSwitchFP, SLGFN, and ThreePhaseFN with a fault-score margin of 0.511616 . Stage 2 performs current-disturbance-based localization using an argmax rule over the installed WMU set. Exhaustive search shows that zero-error zone localization first becomes feasible at $k = 8$, and the selected final placement is $\{2, 11, 13, 16, 20, 23, 24, 27\}$. The study demonstrates that the minimum WMU requirement depends on the monitoring objective: one WMU is sufficient for the evaluated detection task, whereas eight WMUs are required for zero-error zone localization under the specified formulation.

Future work should expand the dataset by varying SSO frequency, SSO amplitude, damping, load level, fault resistance, fault inception angle, measurement noise, missing data, and synchronization error. The hard-constrained framework can also be extended toward train/validation/test thresholding, N-1 sensor contingency constraints, and joint detection-classification-localization placement.

References

- [1] H. Liu, X. Xie, J. He, T. Xu, Z. Yu, C. Wang, and C. Zhang, "Subsynchronous interaction between direct-drive pmsg based wind farms and weak ac networks," *IEEE Transactions on Power Systems*, vol. 32, no. 6, pp. 4708–4720, 2017.

- [2] L. Yuan, K. Meng, J. Huang, and Z. Y. Dong, "Investigating subsynchronous oscillations caused by interactions between pmsg-based wind farms and weak ac systems," *International Journal of Electrical Power & Energy Systems*, vol. 115, p. 105477, 2020.
- [3] W. Liu, X. Xie, H. Liu, and J. He, "Mechanism and characteristic analyses of subsynchronous oscillations caused by the interactions between direct-drive wind turbines and weak ac power systems," *The Journal of Engineering*, vol. 2017, no. 13, pp. 1651–1656, 2017.
- [4] W. Liu, X. Xie, J. Shair, and J. He, "Frequency-coupled impedance model based subsynchronous oscillation analysis for direct-drive wind turbines connected to a weak ac power system," *The Journal of Engineering*, vol. 2019, no. 18, pp. 4841–4846, 2019.
- [5] J. Sun, "Impedance-based stability criterion for grid-connected inverters," *IEEE Transactions on Power Electronics*, vol. 26, no. 11, pp. 3075–3078, 2011.
- [6] B. Wen, D. Dong, D. Boroyevich, R. Burgos, P. Mattavelli, and Z. Shen, "Impedance-based analysis of grid-synchronization stability for three-phase paralleled converters," *IEEE Transactions on Power Electronics*, vol. 31, no. 1, pp. 26–38, 2016.
- [7] M. A. Chowdhury, M. A. Mahmud, W. Shen, and H. R. Pota, "Nonlinear controller design for series-compensated dfbg-based wind farms to mitigate subsynchronous control interaction," *IEEE Transactions on Energy Conversion*, vol. 32, no. 2, pp. 707–719, 2017.
- [8] J. Shair, X. Xie, J. Yang, J. Li, and H. Li, "Adaptive damping control of subsynchronous oscillation in dfbg-based wind farms connected to series-compensated network," *IEEE Transactions on Power Delivery*, vol. 37, no. 2, pp. 1036–1049, 2022.
- [9] S. M. Azimi and S. Lotfifard, "Supplementary controller for inverter-based resources in weak power grids," *IEEE Transactions on Smart Grid*, vol. 13, no. 4, pp. 2886–2896, 2022.
- [10] Y. Yoon, J. Jang, M. Yoon, X. Zhang, and S. Choi, "Deep reinforcement learning-based mitigation of subsynchronous oscillation due to inverter-based resources in weak grids," in *2024 IEEE Industry Applications Society Annual Meeting*, 2024, pp. 1–5.
- [11] M. Izadi and H. Mohsenian-Rad, "Synchronous waveform measurements to locate transient events and incipient faults in power distribution networks," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 4295–4307, 2021.
- [12] —, "A synchronized lissajous-based method to detect and classify events in synchro-waveform measurements in power distribution networks," *IEEE Transactions on Smart Grid*, vol. 13, no. 3, pp. 2170–2184, 2022.
- [13] M. MansourLakouraj, H. Hosseinpour, H. Livani, and M. Benidris, "Waveform measurement unit-based fault location in distribution feeders via short-time matrix pencil method and graph neural network," *IEEE Transactions on Industry Applications*, vol. 59, no. 2, pp. 2661–2670, 2023.
- [14] R. F. Nuqui and A. G. Phadke, "Phasor measurement unit placement techniques for complete and incomplete observability," *IEEE Transactions on Power Delivery*, vol. 20, no. 4, pp. 2381–2388, 2005.
- [15] B. Gou, "Generalized integer linear programming formulation for optimal pmu placement," *IEEE Transactions on Power Systems*, vol. 23, no. 3, pp. 1099–1104, 2008.
- [16] J. Peng, Y. Sun, and H. F. Wang, "Optimal pmu placement for full network observability using tabu search algorithm," *International Journal of Electrical Power & Energy Systems*, vol. 28, no. 4, pp. 223–231, 2006.
- [17] C. Rakpenthai, S. Premrudeepreechacharn, S. Uatrangjit, and N. R. Watson, "An optimal pmu placement method against measurement loss and branch outage," *IEEE Transactions on Power Delivery*, vol. 22, no. 1, pp. 101–107, 2007.
- [18] M. Carvalho, X. Klimentova, and A. Viana, "Observability of power systems with optimal pmu placement," *Computers & Operations Research*, vol. 96, pp. 330–349, 2018.
- [19] M. Eissa and A. Kassem, "Hierarchical clustering based optimal pmu placement for power system fault observability," *Heliyon*, vol. 4, no. 8, p. e00725, 2018.
- [20] S. P. Pokharel and S. Brahma, "Optimal pmu placement for fault location in a power system," in *41st North American Power Symposium*, 2009, pp. 1–5.
- [21] J. Faiz and S. Lotfi-Fard, "Adaptive fuzzy system for discrimination of fault from non-fault switching in over-current protection," *Electric Power Components and Systems*, vol. 35, no. 12, pp. 1367–1384, 2007.
- [22] F. Wang and M. H. J. Bollen, "Classification of component switching transients in the viewpoint of protective relays," *Electric Power Systems Research*, vol. 64, no. 3, pp. 197–207, 2003.
- [23] S. M. Shahrtash and M. Sarlak, "High impedance fault detection using harmonics energy decision tree algorithm," in *2006 International Conference on Power System Technology*, 2006, pp. 1–5.
- [24] N. Perera and A. D. Rajapakse, "Power system transient classification for protection relaying," in *2008 13th International Conference on Harmonics and Quality of Power*, 2008, pp. 1–6.
- [25] F. B. Costa, B. A. de Souza, and N. S. D. Brito, "Detection and classification of transient disturbances in power systems," *IEEE Transactions on Power and Energy*, vol. 130, no. 10, pp. 910–916, 2010.
- [26] N. Hajibandeh, F. Faghihi, H. Ranjbar, and H. Kazari, "Classifications of disturbances using wavelet transform and support vector machine," *Turkish Journal of Electrical Engineering & Computer Sciences*, vol. 25, pp. 832–843, 2017.
- [27] I. Sadeghkhan, M. E. H. Golshan, A. Mehrizi-Sani, J. M. Guerrero, and A. Ketabi, "Transient monitoring function-based fault detection for inverter-interfaced microgrids," *IEEE Transactions on Smart Grid*, 2016.
- [28] P. K. Bera, C. Isik, and V. Kumar, "Discrimination of internal faults and other transients in an interconnected system with power transformers and phase angle regulators," *IEEE Systems Journal*, vol. 15, no. 3, pp. 3450–3461, 2021.
- [29] V. Veerasamy, N. I. A. Wahab, M. L. Othman, S. Padmanaban, K. Sekar, R. Ramachandran, H. Hizam, A. Vinayagam, and M. Z. Islam, "Lstm recurrent neural network classifier for high impedance fault detection in solar pv integrated power system," *IEEE Access*, vol. 9, pp. 32 672–32 687, 2021.
- [30] M. G. M. Zanjani, H. K. Karegar, H. A. Niaki, and M. G. M. Zanjani, "High impedance fault detection of distribution network by phasor measurement units," *Smart Grid and Renewable Energy*, vol. 4, no. 3, pp. 297–305, 2013.